

Outline

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- Network Terminologies
 - Network Topology
 - Links
 - Media
 - Network Protocol
- Network Software
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- Client, Server, Peer

Computer Networks

- Computer networks are systems of interconnected computers and devices that communicate with each other to share resources, information, and services.
- These networks can be as small as a local area network (LAN) within a single building or as large as a global wide area network (WAN) connecting computers across continents.

Uses of Computer Network

- **Communication** : Computer networks enable communication between individuals, organizations, and devices through various means such as email, instant messaging, video conferencing, and Voice over Internet Protocol (VoIP) services.
- **Internet Access** : Networks provide access to the internet, allowing users to browse websites, access online services, and communicate globally.
- **Resource Sharing** : Networks facilitate the sharing of hardware resources such as printers, scanners, and storage devices among multiple users or devices within an organization, reducing costs and improving efficiency.

Uses of Computer Network cont..

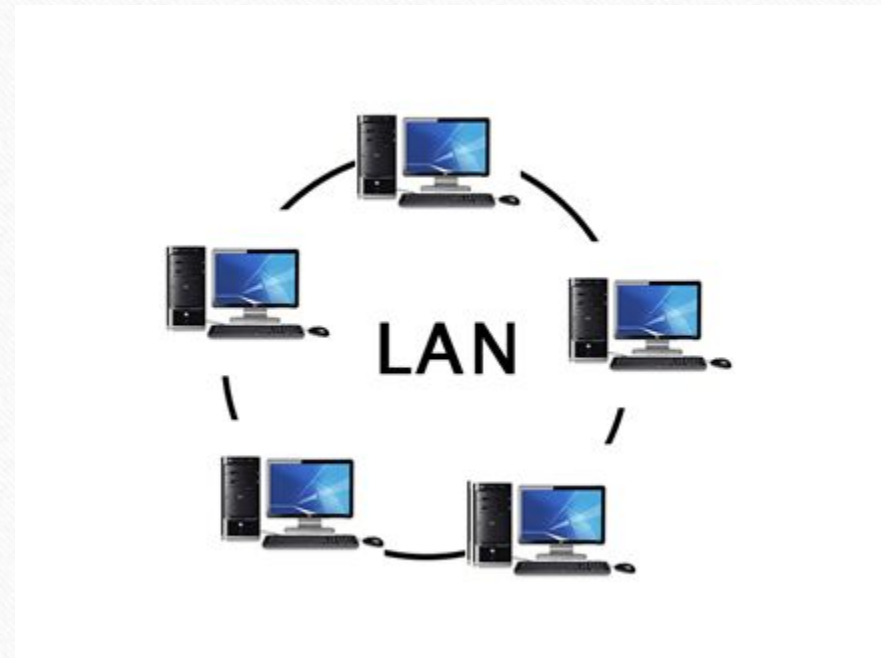
- **Data Storage and Backup** : Networks enable centralized data storage and backup solutions, where data from multiple devices can be stored securely on network-attached storage (NAS) devices or servers.
- **Collaboration** : Networks support collaborative work environments by enabling multiple users to access and edit shared documents, databases, and project management tools in real-time.
- **Remote Access** : Networks allow users to remotely access resources and applications from anywhere with an internet connection, enabling flexible working arrangements and remote management of devices and systems.
- **File Sharing** : Networks facilitate file sharing and transfer between devices using protocols such as File Transfer Protocol (FTP), Network File System (NFS) etc..

Types of Computer Network

- Computer networks can be classified into various types based on their size, geographical scope, and the way they are organized.
- Here are three main types of computer networks:
 - Local Area Network (LAN)
 - Metropolitan Area Network (MAN)
 - Wide Area Network (WAN)

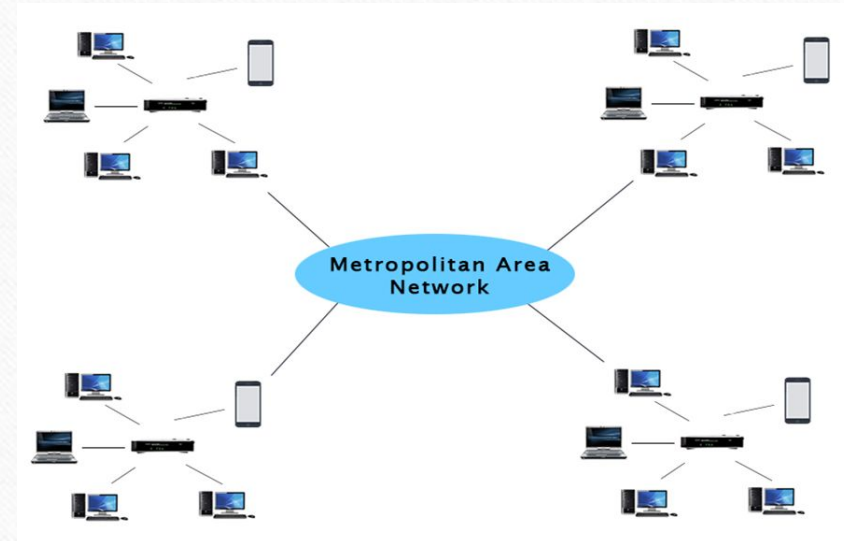
Local Area Network (LAN)

- A LAN is a network that covers a small geographical area, such as a single building, office, or campus.
- LANs typically use high-speed wired technologies like Ethernet or wireless technologies like Wi-Fi.
- They are commonly used in homes, offices, schools, and small businesses to connect devices within a limited area.



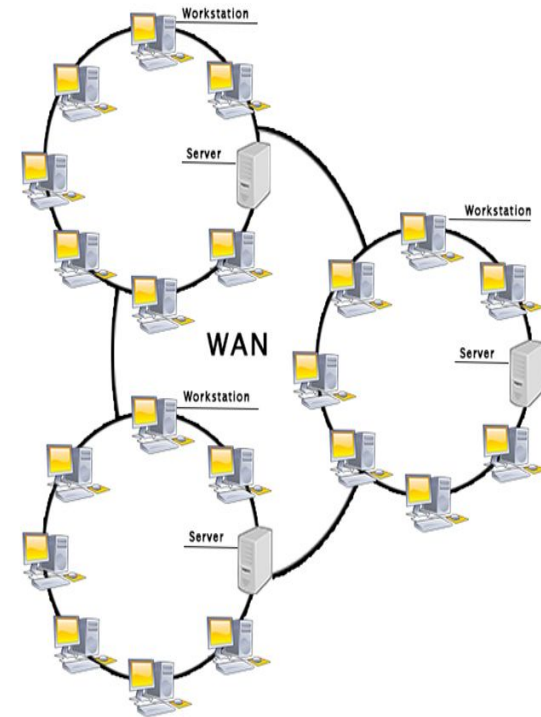
Metropolitan Area Network (MAN)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- It has a higher range than Local Area Network(LAN).



Wide Area Network (WAN)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.



Network Terminology

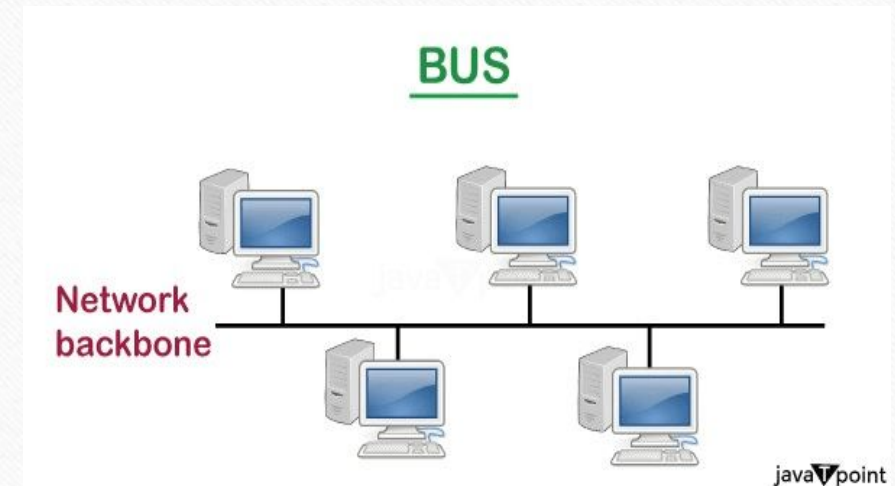
- Network Topology
- Links
- Media
- Network Protocol

Network Topology

- A Network Topology is the arrangement with which computer systems or network devices are connected to each other.
- Topologies may define both physical and logical aspect of the network.
- Both logical and physical topologies could be same or different in a same network.

Network Topology – Bus Topology

- The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- When a node wants to send a message over the network, it puts a message over the network. All the stations available in the network will receive the message whether it has been addressed or not.
- The configuration of a bus topology is quite simpler as compared to other topologies.
- The backbone cable is considered as a "single lane" through which the message is broadcast to all the stations.



Bus Topology cont..

Advantages :

- Requires minimal cable compared to other topologies, making it inexpensive to set up.
- Easy to install, especially for small networks.
- New devices can be added to the bus without disrupting the entire network.
- Reduces dependency on expensive devices like switches or hubs.

Disadvantages :

- If the central bus cable fails, the entire network goes down.
- Identifying the exact point of failure is challenging because all devices share the same cable.
- Adding too many devices can overload the bus and degrade performance.
- Since all devices share the same cable, data collisions are common in large networks, reducing efficiency.
- As the cable length increases, signals weaken, limiting the size of the network.

Network Topology – Ring Topology

- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.
- The data flows in one direction, i.e., it is unidirectional.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a clockwise direction.



Ring Topology cont..

Advantages :

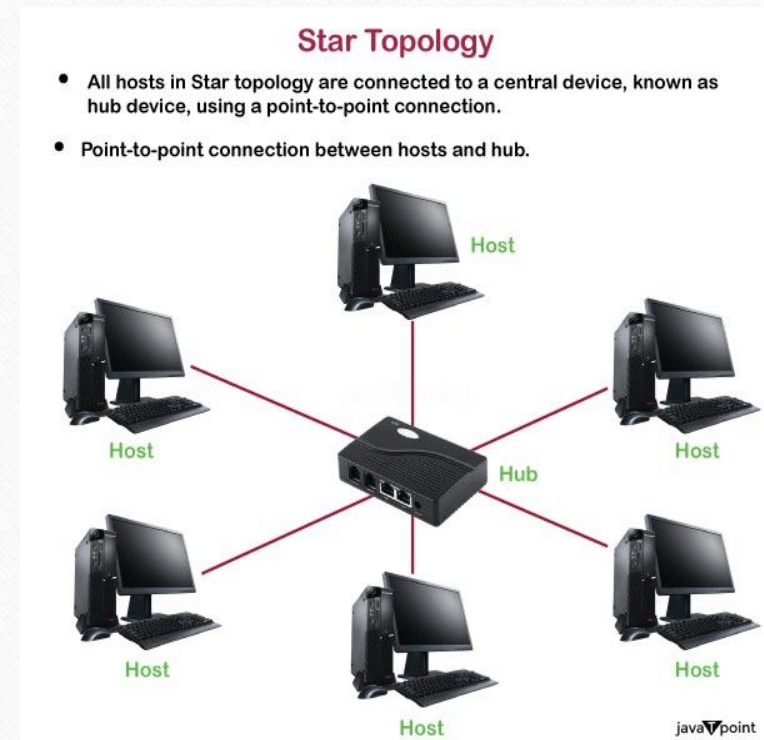
- Data flows in one direction (or two, in a bidirectional ring), reducing the chances of collisions.
- Works well for networks with fewer devices, as data travels through fewer hops.
- Unlike star topology, it doesn't rely on a central hub or switch.
- The time to transfer data is predictable since each device processes data in sequence.

Disadvantages :

- If one device or connection fails, it can disrupt the entire network (unless a dual-ring system is used).
- Identifying the point of failure can be challenging, as the signal passes through each device.
- Adding or removing devices requires temporarily breaking the ring, which disrupts network communication.
- The more devices in the ring, the longer the data takes to reach its destination

Network Topology – Star Topology

- Star topology is a type of network configuration where all devices (nodes) are connected to a central device, such as a hub, switch, or router.
- The hub receives data from a device and forwards it to the intended recipient device.
- If the hub or switch is intelligent (e.g., a network switch), it can direct traffic only to the target device, improving efficiency.



Star Topology cont..

Advantages :

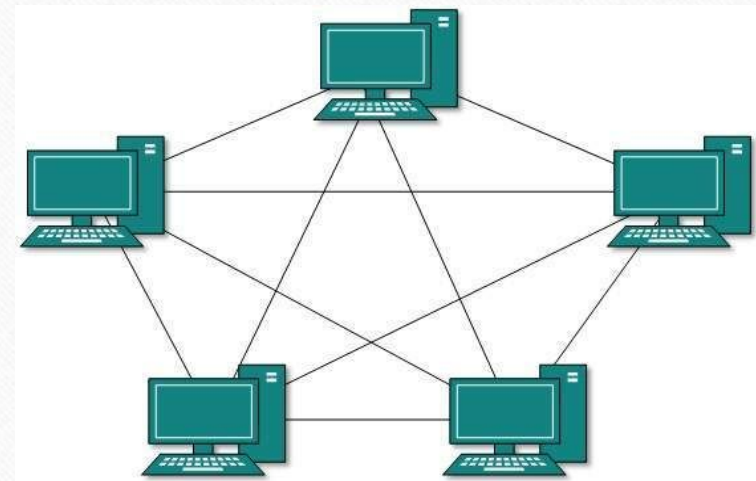
- Simple to set up and configure, especially for small networks.
- If a single cable or device fails, the rest of the network remains unaffected.
- Devices can be easily added or removed without disrupting the network.
- Switches can direct data specifically to the target device, reducing congestion.
- Makes monitoring and troubleshooting simpler.

Disadvantages :

- If the central hub/switch fails, the entire network becomes inoperable.
- Requires more cables (one per device) and a central device, increasing setup costs.
- The performance of the network depends on the capacity of the hub/switch.
- Requires more cabling compared to simpler topologies like bus topology.

Network Topology – Mesh Topology

- Mesh topology is a network configuration where every device (node) is interconnected with every other device, either directly (fully connected mesh) or through multiple paths (partially connected mesh).



Mesh Topology cont..

Advantages :

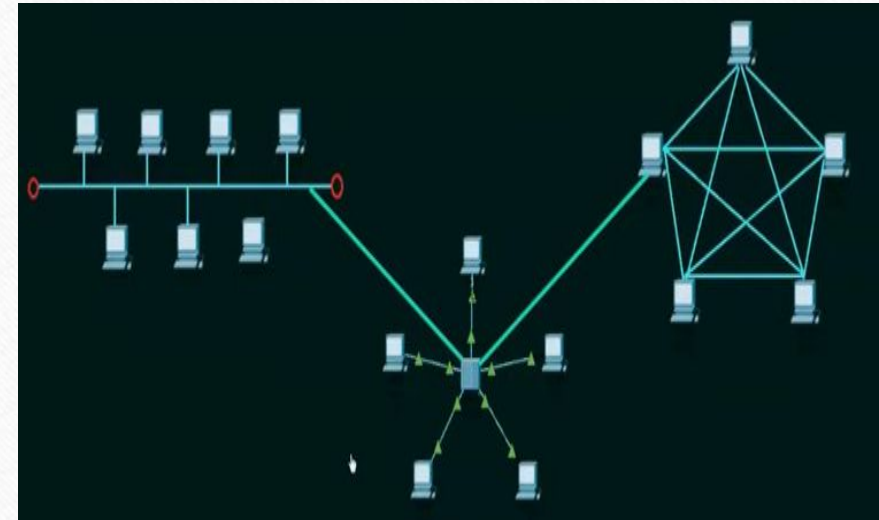
- Multiple redundant paths ensure the network remains operational even if a device or connection fails.
- Since data can travel through alternate routes, the network is more robust compared to star or bus topologies.
- Direct connections between devices can ensure faster data transfer.
- Devices can be added without disrupting the existing network.
- Data travels through dedicated links, reducing the chances of interception.

Disadvantages :

- Requires a large number of cables and network interfaces, especially in a fully connected mesh.
- Many links may remain unused, leading to inefficiency in resource utilization.
- As the number of devices increases, the complexity and cost grow exponentially.

Network Topology – Hybrid Topology

- A network structure whose design contains more than one topology is said to be hybrid topology.
- Hybrid topology inherits merits and demerits of all the incorporating topologies.



Links

- Links refer to the connections between network devices that enable communication.
- These connections can be physical or logical.
- A link defines the path over which data travels between devices in a network.
- Links can be categorized based on various factors, including their physical characteristics (e.g., wired vs. wireless), topology (e.g., point-to-point, multipoint), and transmission speed.
- Examples of links include Ethernet connections between computers and routers, wireless connections between devices and access points etc..

Media

- Media, on the other hand, refers to the physical substances or materials through which data is transmitted between network devices.
- It represents the actual physical medium or channel over which data signals propagate.
- Network media include copper cables (e.g., twisted-pair cables, coaxial cables), fiber optic cables, wireless spectrum (e.g., radio frequencies, light waves), and satellite links.
- Different types of media have distinct characteristics in terms of bandwidth, transmission speed, distance limitations, susceptibility to interference, and cost.

Network Protocol

- Network protocols define rules and conventions for communication between devices in a network.
- They specify how data is formatted, transmitted, routed, and received across networks.
- An example of common protocol architecture is the TCP/IP protocol suite.

Network Software

- Network software refers to a broad category of software applications, protocols, and services designed to facilitate communication, management, and control within computer networks.
- Network software can be categorized into several types based on their functions and purposes:
 - Network Operating Systems (NOS)
 - Network Management Software
 - Network Security Software
 - Network Utilities and Tools
 - Collaboration and Communication Software
 - Network File Sharing and Storage Software

Network Software – Network Operating Systems

- A **Network Operating System (NOS)** is a specialized software designed to manage and coordinate network resources and services for devices connected to a network.
- It provides an environment where multiple computers can share resources, communicate with each other, and work together within a network.
- Examples include Microsoft Windows Server, Linux distributions like Ubuntu Server and CentOS, and Novell NetWare (now deprecated).

Network Software – Network Management Software

- Network management software provides tools and utilities for monitoring, analyzing, configuring, and managing network devices and services.
- Examples include SolarWinds Network Performance Monitor, Cisco Prime Infrastructure, and Nagios.

Network Software – Network Security Software

- Network security software encompasses tools and solutions aimed at protecting networks from unauthorized access, data breaches, malware, and other security threats.
- Examples include firewalls, intrusion detection/prevention systems (IDS/IPS), antivirus software, VPN (Virtual Private Network) clients and servers, and encryption software.

Network Software – Network Utilities and Tools

- Network utilities and tools are software applications used for network troubleshooting, diagnostics, performance optimization, and administration.
- Examples include Ping, Traceroute, nslookup, Wireshark (packet analyzer), Netcat, Nmap (network scanner), and iperf (network performance measurement tool).

Network Software – Collaboration and Communication Software

- Collaboration and communication software facilitate real-time communication, collaboration, and file sharing among users within a network.
- Examples include email servers (e.g., Microsoft Exchange, Postfix), instant messaging/chat applications (e.g., Slack, Microsoft Teams), video conferencing software (e.g., Zoom, Microsoft Teams), and document collaboration platforms (e.g., Google Workspace, Microsoft SharePoint).

Network Software – Network File Sharing and Storage Software

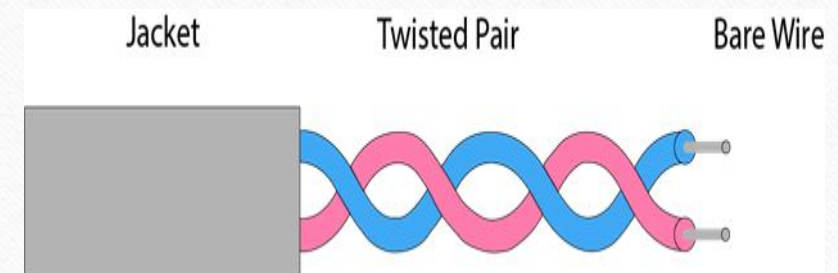
- Network file sharing and storage software enable users to share and access files and resources stored on network-attached storage (NAS) devices or file servers.
- Examples include Network File System (NFS), Server Message Block (SMB)/Common Internet File System (CIFS), and distributed file systems like Hadoop Distributed File System (HDFS) and GlusterFS.

Transmission Media

- Transmission media in networking refer to the physical pathways or channels used to transmit data signals between devices in a network.
- Three main transmission media :
 - Metallic Cable :
 - Twisted Pair Cable
 - Coaxial Cable
 - Optical Fibre
 - Wireless

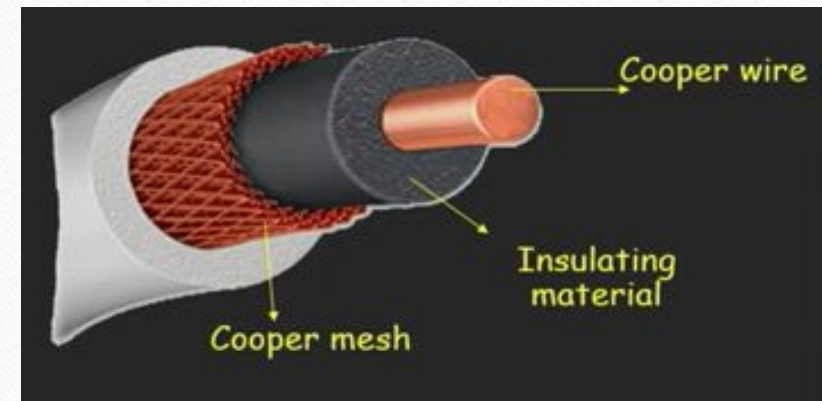
Transmission Media – Twisted Pair Cable

- Twisted pair is a physical media made up of a pair of cables twisted with each other.
- A twisted pair cable is cheap as compared to other transmission media.
- Installation of the twisted pair cable is easy, and it is a lightweight cable.
- The frequency range for twisted pair cable is from 0 to 3.5KHz.
- A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.



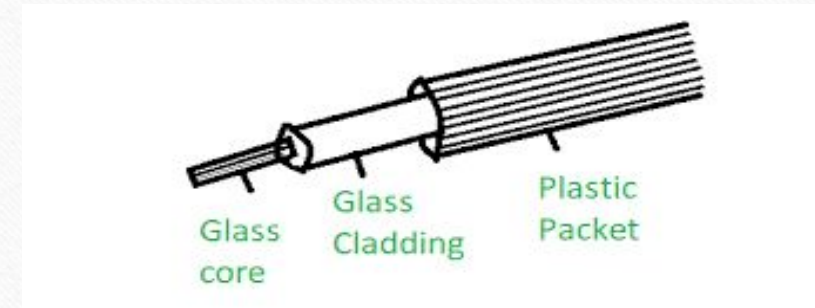
Transmission Media – Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).



Transmission Media – Optical Fiber

- Fiber optic cable, also known as optical fiber cable, is a type of transmission medium used in networking to transmit data signals using light pulses.
- It consists of one or more optical fibers, which are thin strands of glass or plastic that carry data over long distances with high bandwidth and low attenuation (signal loss).
- Main components of fiber optic cable :
 - Core
 - Cladding
 - Jacket



Network Interface Card - NIC

- A network interface card (NIC) is a hardware component without which a computer cannot be connected over a network.
- It is a circuit board installed in a computer that provides a dedicated network connection to the computer.
- It is also called network interface controller, network adapter or LAN adapter.
- NIC allows both wired and wireless communications.
- NIC allows communications between computers connected via local area network (LAN) as well as communications over large-scale network through Internet Protocol (IP).



Client

- A client is a device or software application that requests services or resources from another device or application called a server.
- Clients typically initiate communication with servers to perform tasks such as accessing files, retrieving data from databases, sending emails, browsing web pages, or downloading files.
- Examples of clients include computers, smartphones, tablets, web browsers, email clients, and FTP (File Transfer Protocol) clients.

Server

- A server is a device or software application that provides services, resources, or data to clients in response to their requests.
- Servers are dedicated machines or processes designed to handle multiple client requests simultaneously, serving as central repositories or hosts for shared resources.
- Examples of servers include file servers, web servers, email servers, database servers, print servers, and application servers.

Peer

- In peer-to-peer (P2P) networking, the term "peer" refers to any device or software application that can both request and provide services or resources to other peers within the network.
- Peers in a P2P network have equal status and can act as both clients and servers, sharing resources directly with each other without the need for centralized servers.
- Examples of P2P applications include file-sharing programs (e.g., BitTorrent) etc..

Thank You

